Skin Cancer: Melanoma Classifier

Melanoma ranks as the 5th most common US cancer, with ~100,000 annual cases.

Early diagnosis boosts 5-year survival rate to 99%. Delay leads to worse outcomes.

by Yash Vardhan

The Problem: Diagnosis Challenges

Visual Complexity

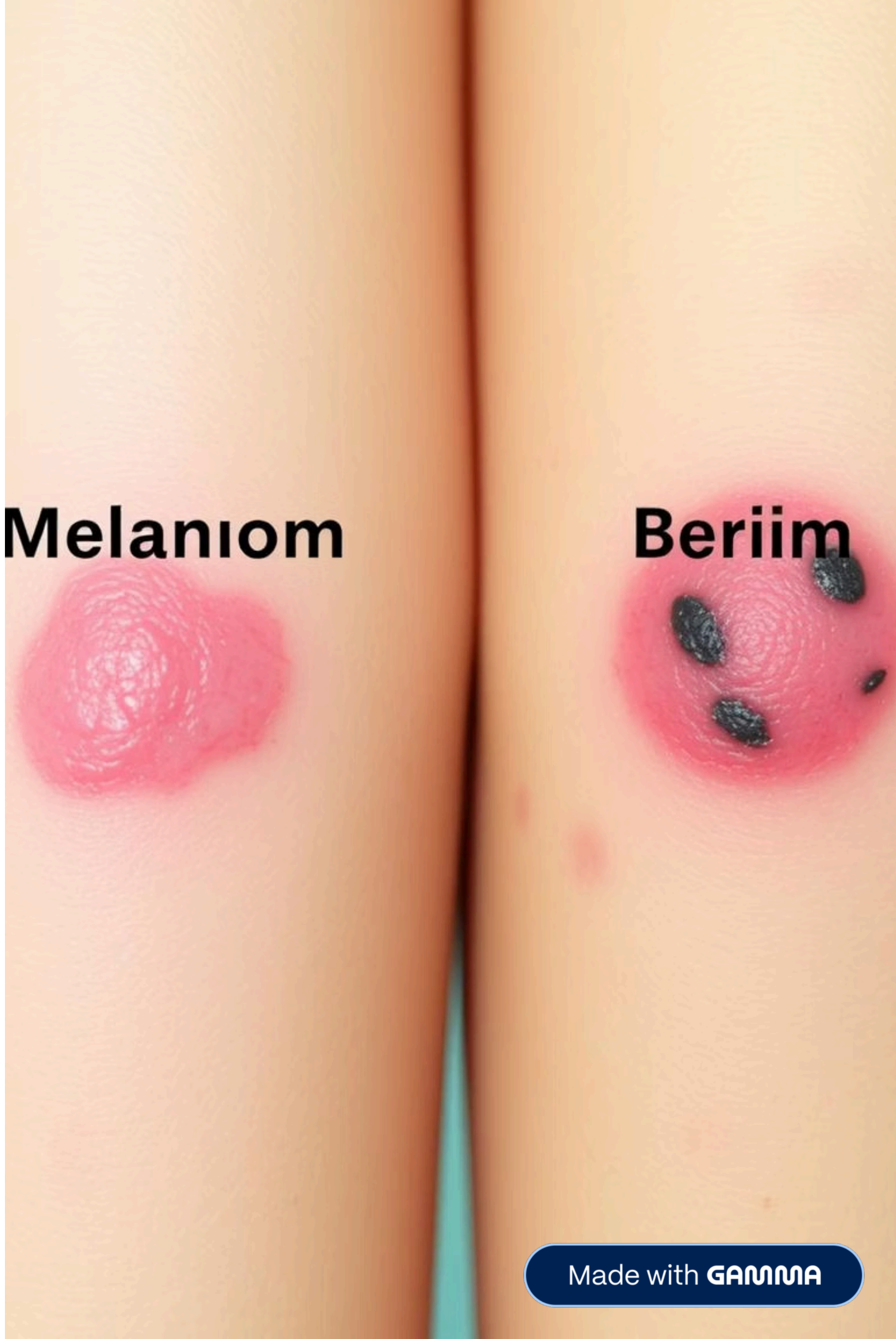
Benign nevi often look like melanoma, causing confusion in diagnosis.

Variability

Dermatologists show 20-30% inconsistency in identifying melanoma from images.

Need

Enhanced accuracy and efficiency are critical to improving patient outcomes.



Our Solution: AI-Powered Classifier

CNN Model

Convolutional Neural Network analyzes dermatoscopic images for melanoma risk.

Training

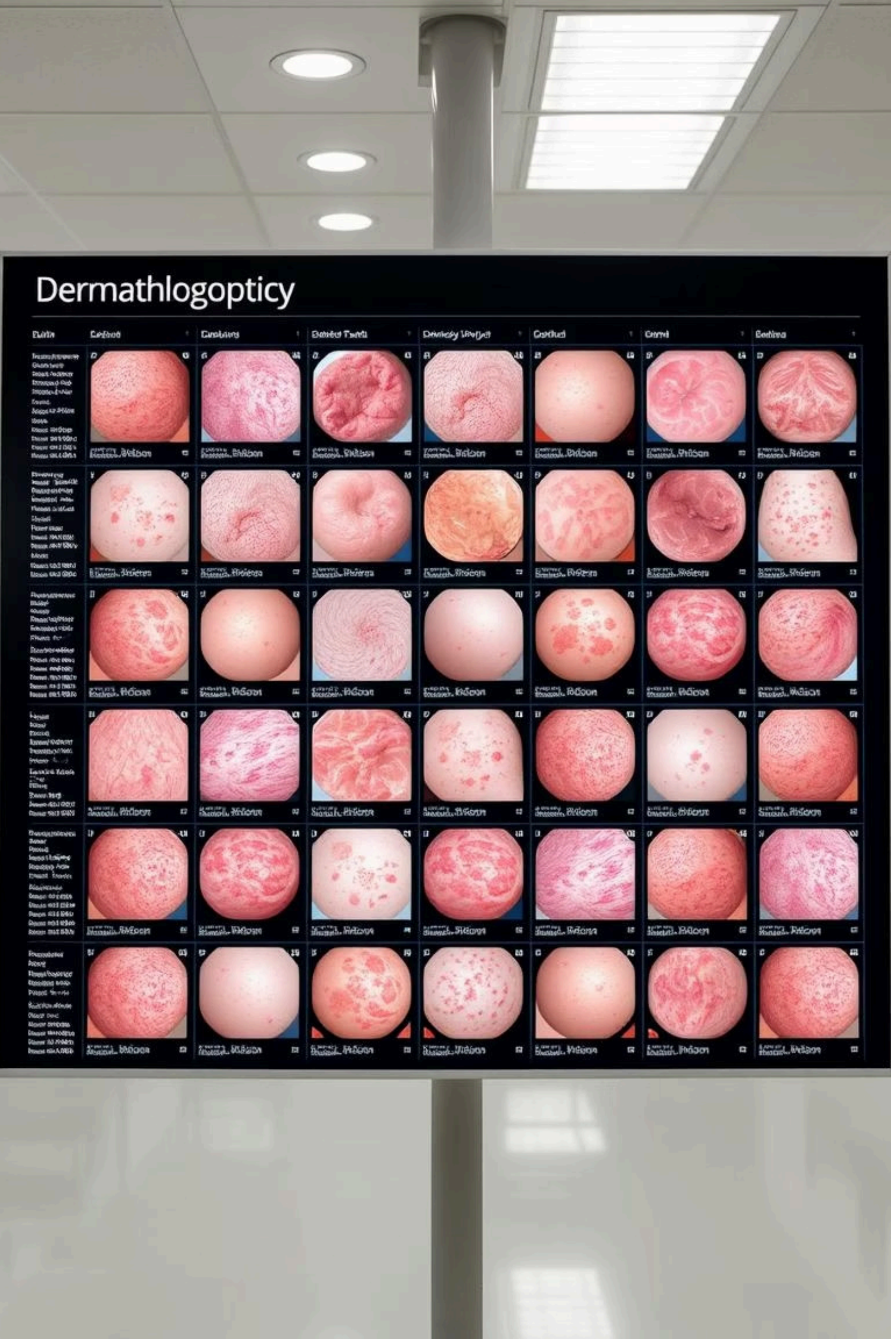
Trained on a large dataset enabling precise skin cancer classification.

Output

Generates probability scores to support clinical decision-making.

Platform

A user-friendly tool assisting dermatologists in diagnosis accuracy.



Data: ISIC Archive & Augmentation

Extensive Dataset

Over 25,000 labeled dermatoscopic images from ISIC archive.

Data Augmentation

Techniques like rotation and zoom increase dataset size by 3 times.

Evaluation

Held-out test set of 5,000 images used for unbiased model testing.

Model Architecture & Training

Architecture

ResNet50 CNN using transfer learning from ImageNet dataset.

Training

Fine-tuned on ISIC data with Adam optimizer and 0.001 learning rate.

Metrics

Accuracy, Sensitivity, Specificity, and AUC used for evaluation.

Results: High Accuracy & Sensitivity

92%

Overall Accuracy

90%

Sensitivity

Effectively detects melanoma cases.

93%

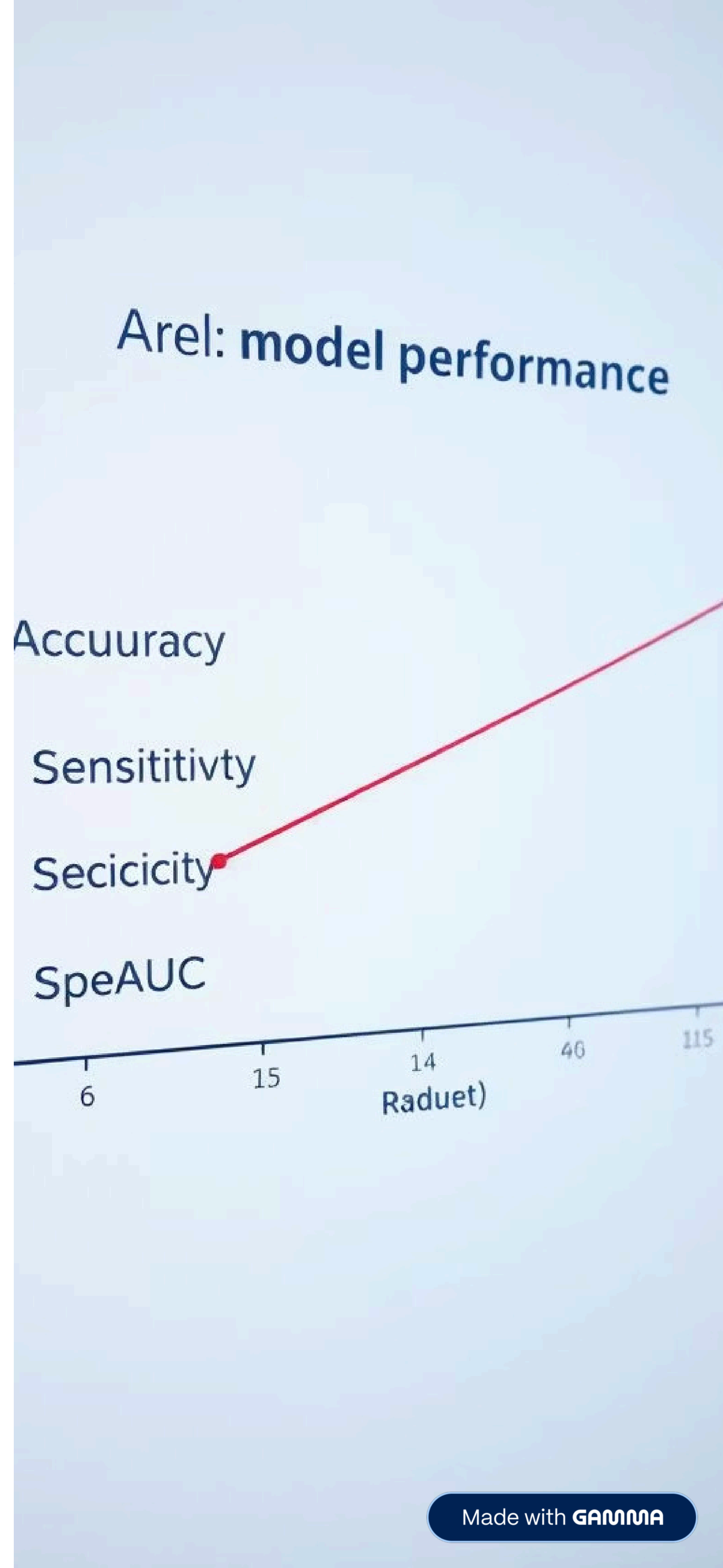
Specificity

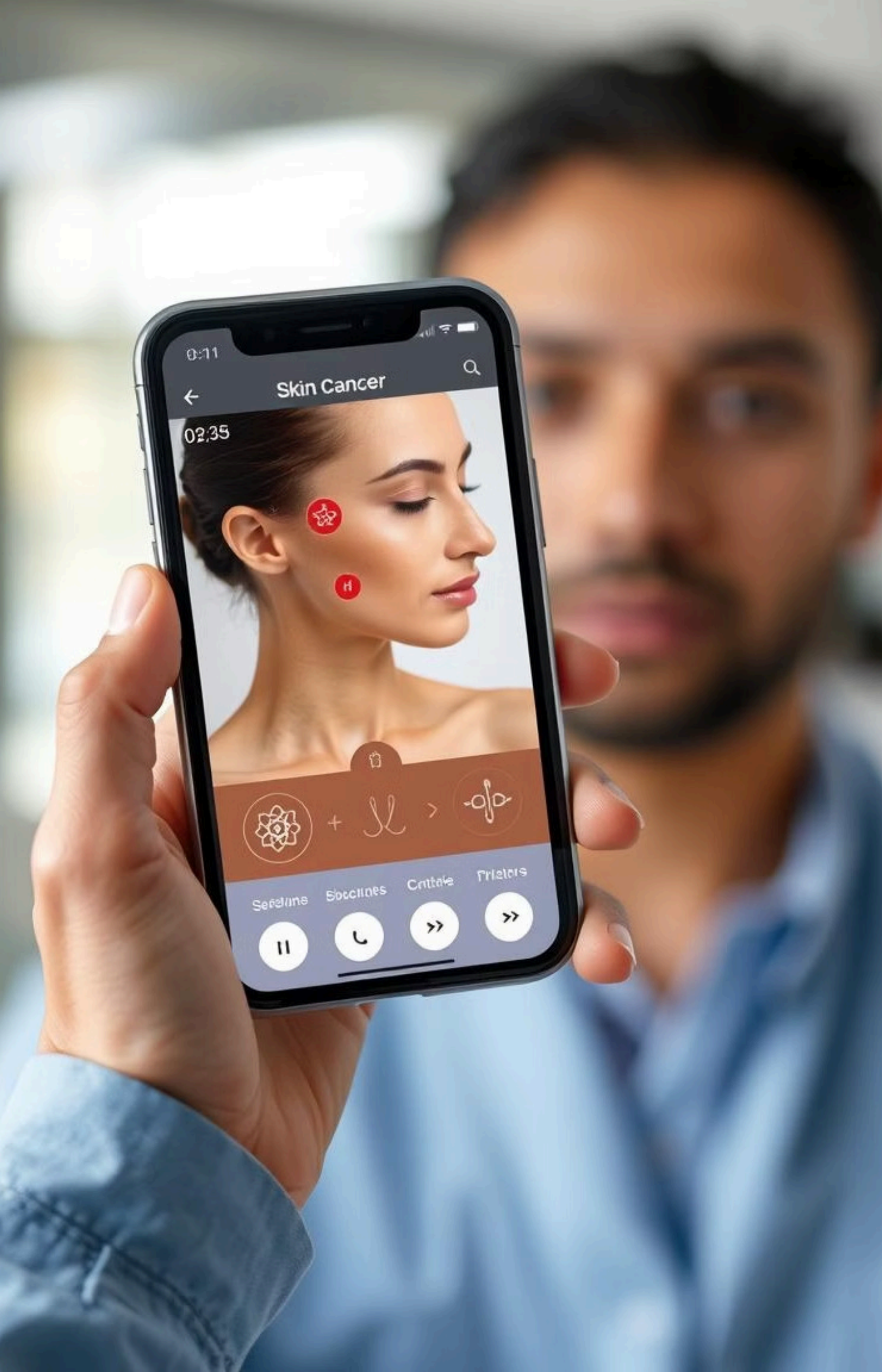
Correctly identifies benign lesions.

0.95

AUC Score

High area under ROC curve indicating strong classification.





Future Work & Improvements

1 Metadata Integration

Add patient info like age and skin type to improve accuracy.

2 Dataset Expansion

Include diverse populations to reduce bias and enhance robustness.

3 Mobile Application

Develop app for real-time point-of-care diagnosis support.

4 Clinical Trials

Conduct validation study with 500 patients for effectiveness proof.

Conclusion: Transforming Melanoma Detection

Accuracy Improvement

AI classifier enhances diagnostic precision and confidence.

Reduced Biopsies

Potential to reduce unnecessary invasive procedures.

Life-Saving Early Detection

Early diagnosis dramatically increases survival rates.

Collaboration Importance

Partnerships with dermatologists essential for clinical adoption.